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Conventional and Unconventional Development Part III: Extrapolation of Conventional Horizontal Fractured Well Development to Lower Permeability Reservoirs

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Abstract

As the world develops ever tighter resources, an often asked question is "when does a conventional hydraulic fracturing approach to field development transition to an unconventional approach." By this we mean when should a viscous fluid, high-density proppant pack stimulation design be replaced by a slickwater fracture fluid design; and also, when should a vertical well development plan transition to a high cluster density horizontal development? The scope of this Part III paper extends the work that was presented in the Part I paper, SPE-223561-MS (Rylance *et al.*, 2025), and the Part II paper, SPE-223562-MS (Pearson *et al.*, 2025). In the Part I and Part II papers the authors discussed the application of multi-stage fractured horizontal wells and presented the potential benefits of a high cluster density, slickwater, multi-stage fractured horizontal well development plan to reservoir permeabilities significantly higher than the nano- and micro-darcy range of typical North American unconventional fields. In this paper, we investigate the use of a horizontal, conventional tight oil & gas development design more typical in millidarcy permeability formations into lower permeability formations and present the results of a study demonstrating the influential factors and driving relationships in such an approach to field development.

The Part I paper outlined some of the issues and experiences related to the intermediate range of mobility between conventional and unconventional fractured developments. This was expanded upon in Part II by considering the direct transposition of an unconventional development approach to low permeability millidarcy and sub-millidarcy reservoirs. In this paper, the intent is to deepen the consideration into identifying the fundamental underlying factors that influence the selection process and to consider the range of cases where the conventional and unconventional completion designs overlap and could be used in field development. A suite of differing horizontal well development approaches and geometries are considered, over a range of permeabilities. This approach allows for greater insight to be gained into the most effective selection criteria and factors influencing development options for low permeability formations. As part of the selection process, the paper attempts to draw on industry completion experiences, both good and bad, and indicate some of the emerging trends. Finally, economics are considered by assigning typical development

costs to each horizontal well approach. This allows the recovery results to be presented as comparative tables of Unit Development Costs (Capital $ spent per BOE recovered) for the differing development approaches as a function of reservoir permeability.

Productivity, recovery, and cost efficiency have been variously presented in a range of graphs; as a function of the reservoir base permeability / mobility. Results indicate that the intermediate range of permeability requires very careful consideration, as the conventional and unconventional production delivery techniques converge. In addition, a number of impactful regionally specific factors are discussed, which may on occasion override, influence and/or directly affect the consideration of the underlying development approach taken and economic results.

With the advent of the development of increasingly tight conventional acreage within existing mature basins, the selection of either conventional or unconventional hydraulic fracturing completion approaches becomes an increasingly important decision. The reported industry experiences have been variable, and it has been the aim of this suite of papers to encourage discourse and generate more detailed and broader publication across such developments. Through the studies presented in this paper the authors provide additional insights which will enhance the industry's efficiency and economics of future low permeability field developments.

INTRODUCTION

As noted within the earlier companion publications, the development of tight-oil and tight-gas are increasingly playing an important role in the global oil and gas delivery portfolio. As these developments are considered and progress into reality, a large part of the debate has turned to when and where to select the application of large-scale conventional or unconventional fracturing practices in field development – especially in the development of low permeability reservoirs. In addition, for many of these non- North American (NA) opportunities, there are other diverse drivers which may impact the field development choices ranging from opportunity limitations due to basin maturity, the need to diversify the energy base, service market conditions, the regional experience base and other market drivers.

By the terminology of a "Conventional" field development approach to horizontal well completions we are meaning the use of horizontal wells with relatively few fracs (compared to an "Unconventional" development). In these cases, the fracture permeability/conductivity will be a dominant factor of productivity and ultimately recovery. By "Unconventional" field development we solely mean horizontal wells completed with high density, tightly spaced fracs where-in fracture conductivity is not as dominant a factor and as such the use of slickwater fluids have been dominant. The wellbore and high-density fractured completion can be more thought of as "mining" the hydrocarbon – we are taking the access path/fracture to the hydrocarbon molecule. Such an approach in North America has enabled the economic development of low microdarcy and nanodarcy formations (themselves referred to as unconventional oil and gas).

In all cases the underlying criteria for the selection of the most effective development approach will include an assessment of the distribution of the formation permeability and variation across the target formation. Simply referring to a formation as ‘tight’ is too subjective to an observer's individual viewpoint and current regional capability, and not necessarily reflective of the underlying fundamental mathematics and economics. In Part I this was described with regard to Figure 1 (Cander, 2012), which indicates that, certainly in a dry gas environment, that an effective conventional planar fracturing treatment window (considered conventional) can be considered applicable in even a low microdarcy (μD) environments.

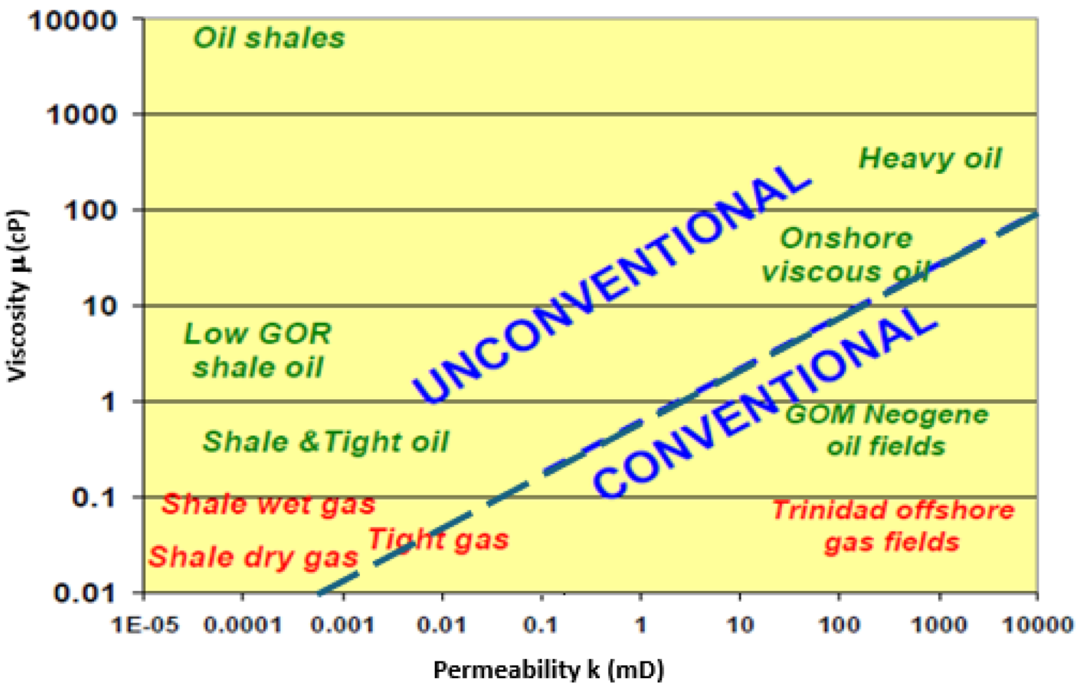

Figure 1—Adapted from Poster Presentation at AAPG Annual Conference and Exhibition (Cander, 2012).

One aspect that was discussed in the Part I paper was that there is comparatively little combined industry experience of conventional, cross-linked, propped hydraulic fracturing (multi-stage) developments. This is in part due to the large volume of unconventional development work in North America (NA) over the past 20-years using the multi-stage, slickwater stimulation design approach. It is also worth noting that at the time of NA low-permeability field development (in the 1970's to 1990's) the technology for multi-stage horizontal well fracturing had not been developed – or was too costly as it required well interventions with a drilling rig as a part of the completions process. But this does not mean that the conventional, cross-linked, propped hydraulic fracturing approach to a low permeability field development is not valid – but rather that it should be evaluated as one of the options for development.

Certainly, where the early industry conventional experience had been gained was typically in higher permeability, oil producing environments - for very logical economic reasons. Such higher permeability formations by their very nature often have a tendency to be a somewhat softer rock (lower Young' Modulus) environment (Tehrani, 1992), (Norris *et al.*, 1996) and (Lietard *et al.*, 1996). This resulted in a certain bias with published experience typically having a combination of low-modulus, moderate to high permeability oil formations. This also resulted in a significant lack of experience in the area of conventional planar fracturing with viscous-gel propped fracs in hard-rock formations.

It is also certainly becoming clearer that a principal driver for the selection of a Conventional approach vs. an Unconventional approach is the relative importance of the pseudo-radial flow behaviour. This is the period of flow after the intra-fracture interference has been achieved; and the production has moved into a pseudo-radial drainage mechanism before finally reaching boundary-dominated flow (with offset wells). It can be surmised that this external drainage mechanism is much more efficiently achieved through a low frac-count, high conductivity, conventional planar fracturing development in moderate permeability formations rather than the converse of high density, low conductivity, unconventional dendritic fracture geometries. But what is the application to low permeability (millidarcy and sub-millidarcy) reservoirs – and when do the two approaches overlap in performance?

This has been investigated by Luo *et al.* (2010) who presented Figure 2 in their investigation of the progressive flow regimes in a multi-stage Bakken formation horizontal field development. The initial bilinear/linear flow period extends into an early radial/elliptical flow period, followed by a formation linear flow period which with time becomes pseudo radial/elliptical before eventually becoming a boundary dominated behaviour.

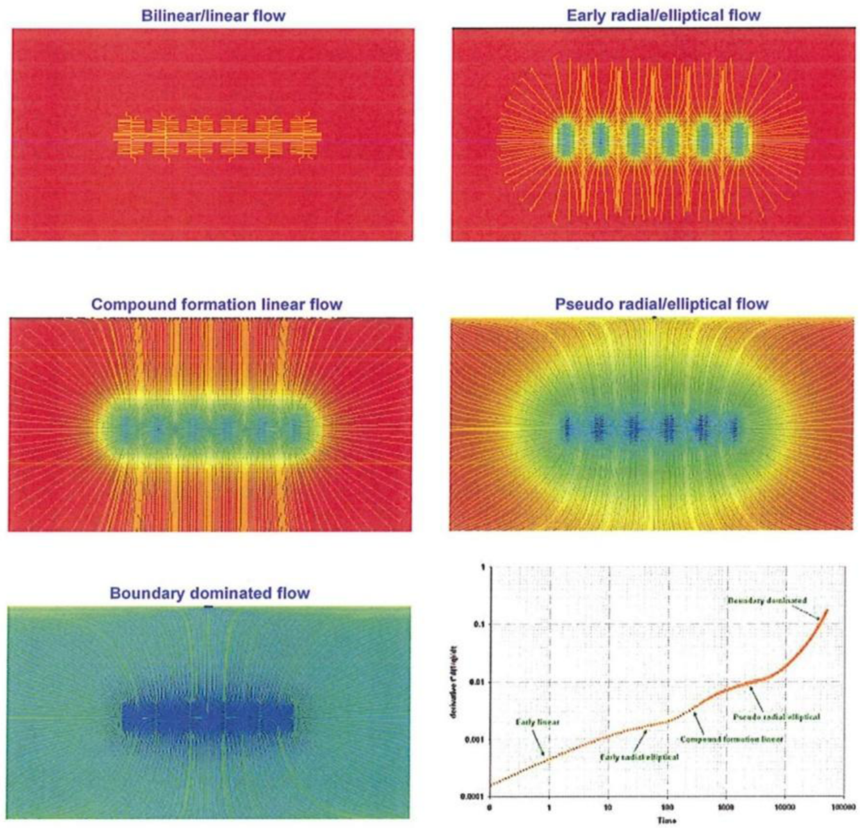

Figure 2—Flow Regimes in an Unconventional Reservoir (after *Luo et al*., 2010).

We can think of this in terms of a Fracture Recovery Ratio (FRR), (Boundary Drainage EUR (%)/ Compound Drainage EUR (%)), where the addition of the Boundary Drainage and the Compound Drainage represent unity. However, their relative value, the FRR, indicates the extent to which Pseudo-Radial Darcy behaviour plays a role in the EUR achieved, and thereby the well spacing and required fracture transmissibility and sustainability. We can think of Conventional fracturing as having a high FRR and conversely Unconventional fracturing as tending to have a low FRR. The importance of this understanding is absolutely key, as this drives well and fracture-spacing requirements as well as fracture geometry and conductivity distribution. Increasingly, the broader industry is beginning to appreciate that while unconventional well and fracture density can (if appropriate) deliver the Compound Drainage, achieving a fully effective Compound Drainage and certainly any major Boundary Drainage requires more effective fracture characteristic consideration (McKimmy *et al.*, 2025).

The intent of this paper is to investigate and discuss the application of a conventional hydraulic fracturing field development approach in a deteriorating quality of rock (i.e. lower permeability) and is fundamentally dual-pronged. The first of these aspects is to perform a comparison of production delivery between conventional and unconventional hydraulic fracturing approaches over a given range of effective permeabilities. This is achieved through modelling/simulation over a broad range of conditions. The second of these aspects is to outline the successful completion approaches that have (and should) be taken in multi-stage conventionally fractured horizontal wells in order to achieve success. Interestingly, it is the second of these aspects that is more challenging for reasons as noted in the Part I paper, which has resulted in somewhat of a publication bias and blind spot across the industry.

These two aspects will now be covered in the subsequent two-sections of the paper.

The Part II paper demonstrated that horizontal wells with multiple transverse fractures are typically capable of producing much more oil than a single vertical well with a single fracture due to the fracture surface area per volume of rock. Furthermore, we also showed that as permeability increases, the relative advantage of horizontal multi-stage fractured wells compared with vertical wells reduces, as would be expected. However, gradual increases in the deliverability of the reservoir (increases in the permeability, porosity, or reduction in the native fluid viscosity and associated flow characteristics) allow for deeper depletion into the reservoir to be achieved during any flow regime (linear, radial, etc.). Additionally, capital costs are directly scaled with the well density and intensity of the completion (number of fractures, volume of proppant and fluid), and there is an optimization opportunity to match the fracturing intensity with the deliverability of the targeted reservoir.

HISTORICAL DEVELOPMENT OF CONVENTIONAL FRACTURING IN HORIZONTALS

North America

The industry's first recorded drilled horizontal well was in Texon, Texas in 1929 and another is recorded in the Franklin Heavy Oil Field of Pennsylvania sometime later in 1944 – but there is little known of the completion and they were likely open-hole completions since they long preceded the industry's development of the hydraulic fracturing process which occurred in the late-40's with its first regional application in the USA in the 1950's and 1960's.

The next recorded development of horizontal wells was in 1979-1982 which were evaluated by ARCO (Atlantic Richfield Company) using a flexible "knuckle-joint" drilling assembly to drill short laterals of 50 to 100 ft. in the Empire Abo Field of New Mexico – a fractured carbonate (Stramp, 1980). Meanwhile, the industry had been developing new technologies for Extended Reach Drilling (ERD) in the development of remote reservoirs in such places as the Gulf of America (GOA), the North Sea, and North Slope of Alaska (ANS). This included technologies such as downhole motors, measurement while drilling, steerable assemblies, and logging while drilling. Not satisfied with the limitations of the "knuckle-joint" drilling technology, ARCO ushered in the advent of modern horizontal development in 1985 utilizing the ERD technologies in the drilling of the John G. Hubbard #1 well into the Austin Chalk formation in Rockwall, Texas. The well was the industry's first medium radius horizontal well drilled with a catenary curve and a maximum 20#/100 ft. build rate in the curve and had 1,500 ft. of lateral extent.

The success of this pilot well ushered in a new phase of industry development across the USA; with the initial application of medium radius horizontal wells being applied to fractured carbonate formations such as the Giddings and Pearsall fields in South Texas. These wells were typically completed with an acid wash or acid frac (Meehan, 1995). In 1989, a total of 257 horizontal well permits were issued in the USA, in 1990 over 1,000 permits were issued, and in 1991 the American Petroleum Institute started tracking horizontal well counts.

Early attempts to combine both horizontal drilling and hydraulic fracturing were not economically successful. In the 1987-88 time period, a number of different companies attempted to combine the technologies in pilot well tests; which were conducted in various low permeability west Texas fields. ARCO's first attempt was with the W. M. Schrock (26) #8 well drilled in the Lower Dean formation with a 2,600 ft. open-hole lateral. The well was then fracture stimulated on July 7th 1987 as depicted in the aerial photograph in Figure 3. The stimulation design was a conventional cross-linked gel treatment with a 1 to 6 PPA proppant ramp and a diverter stage pumped between each of ten individual proppant ramps. The production results achieved were far from spectacular with a maximum production rate increase measured of only ~5 bopd. After a pilot program consisting of a total of three fractured horizontal wells ARCO then abandoned the horizontal well approach to the field development of low permeability reservoirs, as did a number of other companies who had also tested their own horizontal wells in west Texas at the time including both Mobil (White, 1989) and Clayton Williams Energy.

Figure 3—Fracture Stimulation of the W M Schrock (26) #8 well on July 7th 1987.

Mobil's approach documented by White (1989) had made use of a drilling rig to effect zonal isolation between a total of five separate intervals in the lateral – a technique that would yield very positive results when replicated in the North Sea.

Elsewhere in the lower-48, the largest onshore use of horizontal drilling with hydraulic fracturing was to develop the diatomite in the Belridge Field of California. Between 1995 and 2010 a total of 246 horizontal wells were drilled primarily for recovery of oil from the flanks of the field. Wells were drilled in a combination of directions that were longitudinal, transverse and oblique to the preferred stress orientation with lateral lengths from 500 to 3,000 ft. The permeability of the diatomite ranges from 0.1 to 10 mD, and stimulation of the laterals were typically carried out with 4 to 8 stages of conventional cross-linked gel and sand proppant treatments (Wright *et al.*, 1997 and Allan *et al.*, 2010).

North Sea

In 1987, the approach of using rig-isolated fracture stages would be developed further in the re-development of the Dan field in the Danish North Sea. As documented by Kogsbøll *et al.* (1993) and shown in Figure 4, a horizontal well development was designed incorporating 14 separate, drill-rig deployed isolated stimulation stages.

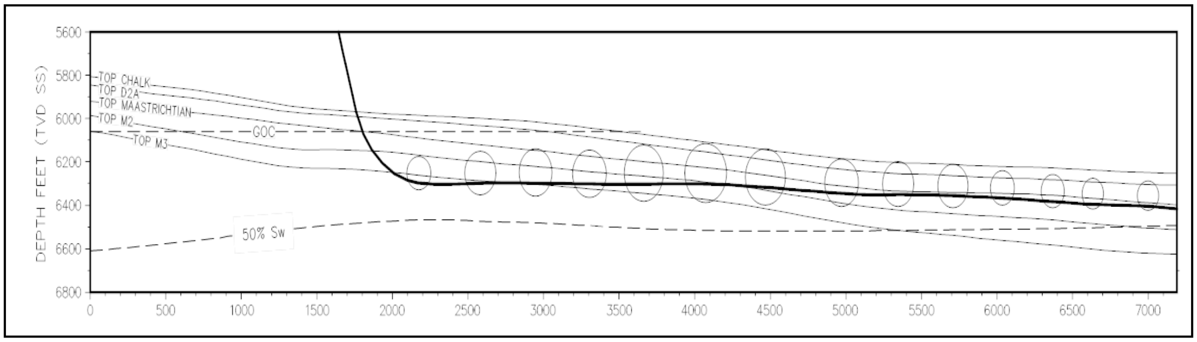

Figure 4—Schematic of the 1987 Dan Field Horizontal Well Pilot Test Wells (Kogsbøll *et al*., 1993).

A two well pilot was implemented in 1987 with spectacular production results, and a tripling of field production was achieved with the addition of ~20,000 bopd, see Figure 5. Further horizontal well, tool and technique development continued with combined field production then achieving a maximum of over 120,000 bopd in the early 2000's. With time, as poorer and thinner flank reservoir was developed, for cost and efficiency reasons completions were adapted to uncemented liners with pre-drilled holes for acid jetting of the soft chalk formation. However, the genie was out of the bottle and the race to develop fields with fractured horizontals began.

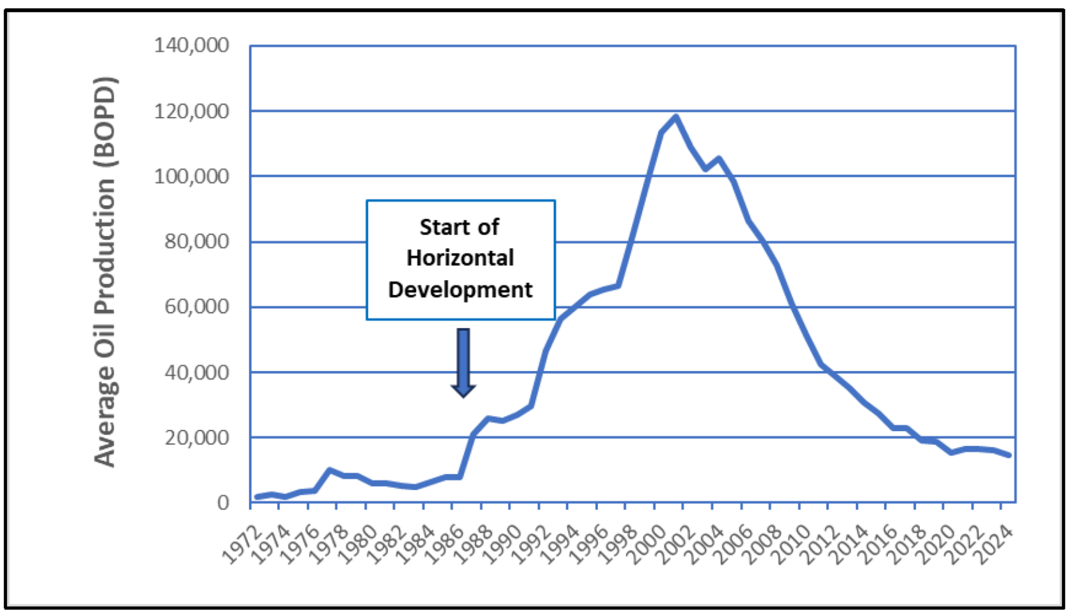

Figure 5—Dan Field Production History.

In 1989, the concept of using horizontal wells for thin reservoir development was used for North Sea development (Brekke *et al.*, 1993). Elsewhere in the North Sea, as well as onshore Europe, conventional hydraulic fracture treatments of ERD wells became much more routine in such formations as the tight-gas Rotliegend (Chambers *et al.*, 1995 and Abou-Sayed *et al.*, 1996), as well as the soft-chalks of the Norwegian Continental Shelf (NCS) (Norris *et al.*, 1998).

The use of fractured horizontal wells really took off in the North Sea for the flank development of the Valhall field in the early-2000's. The operator, (BP Norway), evaluated horizontal wells with a range of different completion types: acid fracture treatments and high conductivity conventional propped fractures. The conventional proppant fracture treatments were found to perform much better than acid fracturing as shown in Figure 6 (Olson *et al*., 2003).

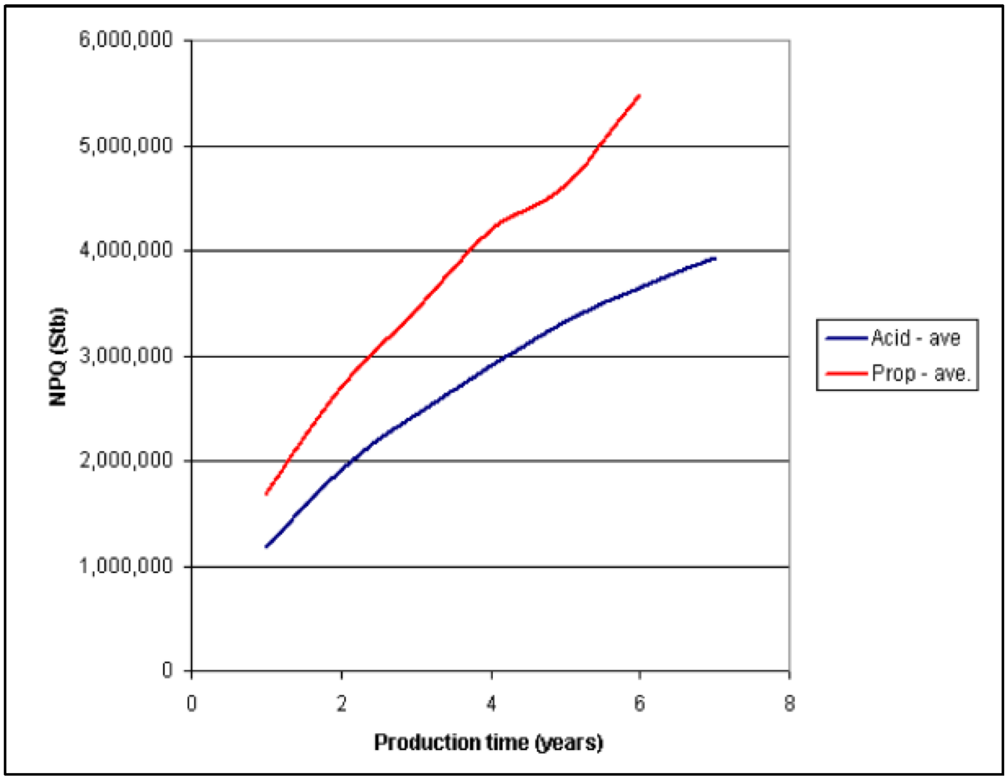

Figure 6—Horizontal Well Performance of Acid and Proppant Fractured Completions in the Valhall Field (Olson *et al.*, 2003).

European onshore and North Sea horizontal well field developments have continued apace, particularly within the Rotliegend Formation, although with a variable frequency and very often consisting of no more than between 1 – 5 wells in total. This is particularly true of the German (Abou-Sayed *et al.*, 1996), (Ehrl & Schueler, 2000), Dutch (Baumgartner *et al.*, 1993), (Schrama *et al.*, 2011) and UK Southern North Sea (SNS) sectors. Consisting of multi-stage (both Transverse and Longitudinal) fractured horizontal wells, cased-hole and open-hole approaches have both been applied, with frac stage count rarely being above a single-digit value. Fracturing has almost exclusively utilized conventional cross-linked gel fluids (principally driven by the fluid leak-off and viscosity requirements) with only a rare and usually unsuccessful excursion from this baseline. In the UK SNS numerous small gas fields such as Babbage (Al-Shamma *et al.*, 2014), Caister (Langford *et al*., 2014), Chiswick (Norris *et al.*, 2022) and Clipper South (Shaoul *et al., 2013*) all fall within this description.

As noted in the Part I paper, the combination of hard rock, horizontal wells, cased-cemented and plug and perforation approach has proven to be adequate, but has not really excelled, e.g., unpredictable PLT contribution (Langford *et al*., 2014) and similarly (Ishteiwy *et al.*, 2017). There may be several reasons for this underperformance, some of which are known and fully appreciated (e.g. the timing between stages) and others which are likely less fully understood even (e.g. complex/poor near wellbore connection based on late events in the pumping and closure, particularly if no screen-out occurs). Combine with this an inability to effectively treat subsea wells, largely due to Rig Well Control policies and concerns about extensive production of proppant post fracture into subsea systems, even with the widespread use of Resin-Coated Proppant (RCP). The result was that by 2015, the qualifying opportunities were reducing and the fracturing of horizontal wells, both dry-tree and proposed subsea were drying-up.

This prompted an Operator Aker-bp to bring in CT deployed/sliding-sleeve solutions in 2015 (Hoq *et al.*, 2022), which was later extended to various pipe-deployed fracturing approaches, either from existing platform structures (Bird *et al.*, 2023), heavy-duty jack-ups (Thomson *et al.*, 2022) and (Young et al., 2022) and even subsea deployments, with a view to implementations with Semi-Sub Rigs in subsea environments as required. The results have been extremely effective, for all of the following reasons:

i. The application of CT or Jointed-Pipe deployed fracturing with sleeves, allows an Operator to move from the execution of a conventional hydraulic fracture treatment every 2 to 3 days, to execution of a conventional hydraulic fracture treatment 2 to 3 times a day. Even in reasonably low stage count fractured horizontal wells, such time reductions have a massive impact on the overall cost and efficiency;
ii. The application of CT or Jointed-Pipe deployed fracturing with sleeves, allows an Operator to purposely allow and design for a screen-out at the end of a treatment, significantly enhancing the quality of the near-wellbore connection. Additionally, this ability to screen-out and continue (reversing or circulating) eliminates clean-out risk; provides for a seamless execution process and provides well control assurance.
iii. The application of CT or Jointed-Pipe deployed fracturing with sleeves (re-closeable), allows for each fracture to be isolated after each stage, and pressure-testing or should suspension of the well is necessary. This is also the case at the end of the entire fracturing stage sequence when the drilling unit will depart/move from the well and the well components are fully tested to API 19AC V0 (ISO 14998) requirements particularly helpful for subsea.
iv. The application of CT or Jointed-Pipe deployed fracturing with sleeves (re-closeable), allows for re-entry at any time and for selected sleeves to be opened or closed. As experience is gained with such practice this may become a significant assist in terms of providing the most economic offtake, particularly in the case of undesirable phases such as water or excessive GOR in oil wells.
v. The application of CT or Jointed-Pipe deployed fracturing with sleeves (re-closeable), allows an Operator to immediately isolate the near-wellbore region and ‘protect’, what is in effect a miniscule connection path often less than 2 – 3 in^2. With conventional approaches, milling and overbalance operations in a well with such finite connections almost certainly risks damaging one or more of the stages.
vi. The application of CT or Jointed-Pipe deployed fracturing with sleeves (with a screen position) allows an operator to select a screen-size such that upon production all the sleeves may be moved to the screen-position, which then provides full proppant retention behind-pipe in the formation (as well as removing RCP complexity and cost). This is a particularly helpful for subsea environments for flow assurance.

Some operators, such as Aker-bp, have been deploying these techniques for nearly 10 years now, in both Cased-Hole and Open-Hole environments. Others are beginning to apply as they develop their own solutions and as the industry gains deeper confidence and track-record in available tools. The result for all parties is that increasingly there are new field development opportunities emerging that had previously been considered too challenging. Certainly, from the mechanical completion perspective it would appear that the right options now exist for such operations. However, for fracture design, the fluids and the materials that are used remains an area that further work and transposition from Unconventional needs to be considered and progressed.

North Slope of Alaska (ANS)

In the mid-1990's in Alaska, at the Kuparuk River Field, the operator (ARCO Alaska) decided to test longitudinally fractured horizontal wells, drilled in the direction of the minimum principal stress. Laterals of 2,600 ft. extent were drilled and completed with one to three stage conventional, crosslinked gel, high proppant loading stimulation treatments (Pearson *et al.*, 1996). The results were a ~35% uplift in performance compared to offset conventionally fractured deviated wells as shown in Figure 7 and paved the way for later "end-of-life" field development of peripheral pads with multi-stage fractured wells and side track multi-lateral horizontal wells drilled with a dedicated Coiled Tubing Drilling rig (Jensen *et al.*, 2012). This performance and regional experience base also provided the know-how for multiple smaller ANS fields to move into horizontal fractured well development such as the Colville Sands at the Alpine Field (Schneider *et al.*, 2007) and the Nuiqsut Formation at the Oooguruk Field (Bond *et al.*, 2015).

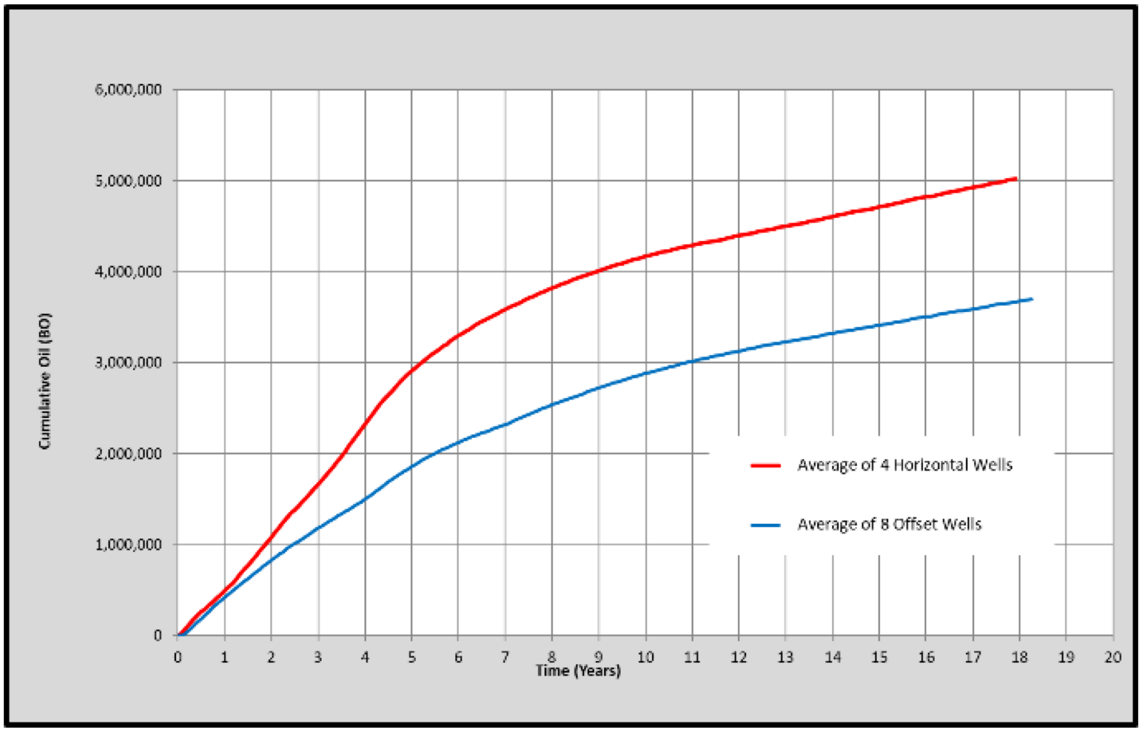

Figure 7—Initial Kuparuk Horizontal Well Performance Compared to ERD Offset Wells (Pearson *et al.*, 2013).

Northern Russia

Another large Oil and Gas Region that significantly benefitted from horizontal drilling during the 1990s was the Western Siberian slope (Jelsma *et al.*, 1993) equivalent in more ways than one to the North Slope in Alaska. While the early well completions were largely limited due to the poorly considered use of slotted liners, as with early NA wells, it quickly became apparent that such completions were neither effective nor flexible as an approach. This scenario then changed considerably, post glasnost, with a number of innovative companies such as YUKOS (Pongratz *et al.*, 2008) and Tnk-BP (Martin and Rylance, 2010), who quickly transferred the latest learning into their horizontal wells. The result was an extensive redevelopment of the great Western Siberian mature fields and a significant rejuvenation and uplift of the resulting Western Siberian production levels, through the drilling of multi-fractured horizontal wells (Dedurin *et al.*, 2006) and (Soliman *et al.*, 2006).

These multi-fractured horizontal wells were usually completed with conventional propped hydraulic fractures; with a fracture count typically between 4 – 8 treatments, over some 500 – 1,000m lateral length. Indeed, Western Siberia is likely the single largest basin, field, well-count and set of experiences with multi-stage conventional propped hydraulic fracturing from horizontal wells. For this reason, it is extremely interesting to note that the vast majority of these wells are completed with open-hole completions, along with some form of external isolation mechanism. This combination was arrived at for a number of key reasons, including the following points:

i. Ease of the initial fracture initiation, low-pressure, simplistic and highly repeatable/uniform behaviour, and

ii. Evidence from Production Logging Tools (PLT) of even stimulation distribution effect (Ishteiwy *et al.*, 2017), and
iii. The ability to flush treatments reasonably accurately without requiring an extensive over-flush or a CT clean-out.

Northern Oman Salt Basins

As noted in the Part I paper, the Fahud and Ghaba salt basins of Northern Oman (Thomas *et al.*, 2015) have been extensively developed, over many decades. Starting of course in the shallow high permeability oil horizons, and gradually developing the deeper and deeper resources, with less liquids and ultimately dry gas. As a microcosm of increasingly difficult reservoir development, this area of Northern Oman provides a useful, and helpful viewpoint and understanding of where the challenges lie, particularly with respect to well design and hydraulic fracturing considerations.

In addition to the fact that there are numerous producing formations within these salt-basins, the very ancient nature of the basin (485 Ma) means that some of the oldest (geologically) producing fields in the world exist there. This age also means that the structures have undergone extensive and varied tectonic history, leading to a robust strike-slip stress-state across the region. With production from the shallower higher permeability formations having begun in the 1950s (Heward *et al.*, 2024), there is an extensive knowledge suite related to the geology down to 5,000m. As the shallow potential matured, deeper and tighter formations have been appraised for their production potential (Rylance *et al.*, 2011a). Inevitably, during the last 30 years, this has led to an increasing consideration of the technology of hydraulic fracturing, both conventional and unconventional, vertical, and horizontal wells.

Conventionally Fractured Verticals (Tight-Rock). Fractured vertical wells have been part of the basin(s) development history since the tighter resources were discovered, with one of the more extensive formations being the Barik. Early field developments such as the Barik, Saih Nihayda Barik and Saih Rawl (Jones *et al.*, 1998) and (Langedijk *et al.*, 2000) paved the way with multi-stacked fracture treatments in the verticals, with later developments such as Khazzan-Ghazeer extending this into massive hydraulic fracturing (Rylance *et al.*, 2011b) where appropriate. Khazzan-Ghazeer rock quality was likely sub-millidarcy on average (0.1 – 0.5 mD) and as such was developable with fractured verticals where sufficient thickness to deliver economic kh existed. However, decisions over single-stage or multi-stage verticals or consideration of horizontals is a question that was revisited multiple times (Parada *et al.*, 2023).

Fundamentally, the use of conventionally fractured verticals is possible, where the well construction costs can be constrained and in the specific case of the Khazzan-Ghazeer Barik formations, for this permeability range, where an economic cut-off of 15 mD.ft of pay could be encountered. Other formations within the same Middle Cambrian to Lower Ordovician clastic reservoirs of the Haima Supergroup have yet to be economically developed.

Conventionally Fractured Horizontals (Tighter-Rock). Numerous investigations and extensive trials have been performed with conventionally fractured horizontals in these basins within the tighter permeabilities that cannot be developed with fractured vertical wells (Nicolaysen *et al.*, 2016) and (Shaoul *et al.*, 2022). However, multiple factors intersect to make these wells less than economic, principally related to drilling costs, these are on average 150-day wells to drill, but also deliverability/uncertainty. As noted in prior publications (Rylance, 2024) the track record of conventional fracturing in cased-cemented horizontals, in harder rock formations is less than stellar. However, as noted in this basin (Ishteiwy *et al.*, 2017) and (Al-Shueili *et al.*, 2022), the use of open-hole with mechanical isolation appears to address this, further confirming that it is the nature of the near-wellbore connection that is dominating.

No doubt the debate about the potential use of conventional fractured horizontals will continue (Parada *et al.*, 2023), but while underlying well construction cost(s) and timing(s) remain as high as they are, it is unlikely that much further field development will take place with this approach (within these specific basins).

Unconventionally Fractured Horizontals (Tightest Rock). A number of specific trials have also been performed within these much tighter formations, where more unconventional approaches have been deployed (Clark *et al*., 2013), (Briner *et al*., 2015), (Ishteiwy *et al*., 2018) and (Abdulkadir *et al*., 2023). While the industry has an unfortunate tendency to only publish success, even so it is clear that proppant placement is a success measure utilized by many, whereas well/development economics are truly the underlying driver for operators. While many of these publications indicate some successful placement, you can be forgiven for thinking that this has been the solution that has subsequently been applied and that the issue is now concluded. However, there is little recorded production history success published; or indeed any other associated documentation and subsequent development behaviours suggest generally poor well deliverability was more widely achieved than has been acknowledged (Elliott & Al-Abri, 2015).

In too many cases, the underlying obsession appears to have been that a specific fracturing technique will be the magic-bullet to the extremely tight nature of the rock, rather than an engineered holistic solution that actually considers the fundamentals of the underlying productivity and economics. No amount of effective stimulation will overcome a 150-day drilling duration, or make unconventional well density affordable, under these specific conditions.

PRODUCTIVITY MODELING

In order to develop an understanding of the potential production performance of a conventional fracturing approach to lower permeability formations, the authors setup a simulation matrix, in which a base case high-conductivity proppant pack completion design was modelled for three different development scenarios beginning with a moderate reservoir permeability of 45 millidarcy:

- A vertical well in a 640-acre spacing unit,
- A horizontal well that has been drilled in the direction of the ***maximum*** horizontal stress with a longitudinal, high conductivity conventional fracture, and
- A horizontal well that has been drilled in the direction of the ***minimum*** horizontal stress and thus having a transverse fracture system with a single fracture each 500 ft. of lateral stage length.

For all three cases, the base simulation details run for each of these cases can be seen in Table 1 below:

Table 1—Horizontal Well Stimulation Matrix.

Parameter	Value	Units
Stage length	500	ft
Fractures per stage	1	-
Fracture height	300	ft
Fracture half-length	375	ft
Base reservoir permeability	45	mD
Fracture conductivity *	5,000	mD.ft

* Fracture conductivity is reduced as a function of effective normal stress as exp(-1*0.0005*effective normal-stress).

For consideration of the fracture properties, while many "soft" formations such as the extensive chalks in the North Sea require the placement of high proppant loading fractures of > 4 lb/ft^2 – for this simulation matrix as we are considering "hard rock" fracturing placement, so we chose a more conventional proppant loading of 2 lb/ft^2 using a premium Light Weight Ceramic (LWC) proppant. As shown in Figure 8, for the case of ~8,000 psi effective stress this loading of either 20/40 or 16/20 LWC proppant yields a base fracture conductivity of ~5,000 mD.ft which was used in the reservoir simulation. While selecting this value of proppant conductivity it is fully recognized that this will only offer infinite-acting conductivity behaviour (i.e. with Dimensionless Fracture Conductivity values >10) at permeabilities of 1 mD and below.

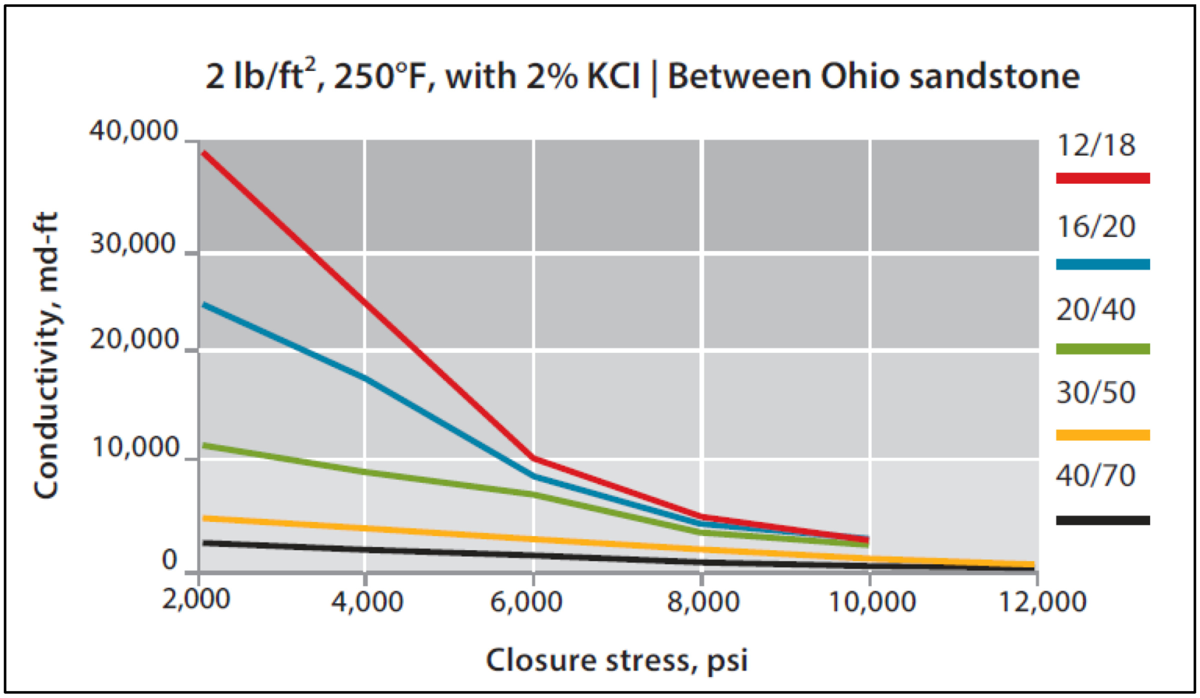

Figure 8—Conductivity of 2 lb/ft² Premium Light Weight Ceramic Proppant.

Reservoir properties for the simulation are provided in Table 2 below.

Table 2—Reservoir Properties for the Horizontal Well Simulation Matrix.

Characteristic	Unit	1	2	3	4	5	6	7
Pay Zone Thickness	(ft)	50	50	50	50	50	50	50
Permeability	(mD)	0.000045	0.00045	0.0045	0.045	0.45	4.5	45.0
Porosity	(%)	6.5	11.5	20.4	25.0	25.0	25.0	25.0
Pore-Pressure	(psi)	4800	4800	4800	4800	4800	4800	4800
Water Saturation	(%)	25.0	25.0	25.0	25.0	25.0	25.0	25.0
Initial R_s	(mcf/bbl)	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Bubble Point	(psi)	650	650	650	650	650	650	650
Oil Viscosity	(cP)	0.5	0.5	0.5	0.5	0.5	0.5	0.5

To investigate the applicability of this conventional style of planar proppant fractured horizontal completion, to other lower permeability reservoir environments, the models were re-run with two ranges of sensitivity: A series of 10-fold reductions in the permeability (with porosity also decreased logarithmically as a function of the permeability). As well as a range of variable fracture spacing for the Transverse fractured completion case from just one hydraulic fracture per stage (i.e. 500 ft separation between fracs in the well) to 2, 5, and 10 fracture locations per stage. (i.e. representing a completion with a perforation cluster spacing of 250 ft, 100 ft, or 50 ft in each stage interval). For all these cases the simulations were performed to cover a 30-year production period.

MODELING RESULTS

In order to make an effective direct comparison of the production and recovery results, we considered the simple case of a 1 square mile (640-acre) development spacing unit. Figure 9 shows the 30-year production profiles of the base 45 millidarcy moderate permeability case for the three reservoir completion designs being considered; assuming that the horizontal well cases are 5,000 ft in lateral length (i.e. having ten 500 ft stages). As expected in such a high permeability environment, the cumulative recovery is the same for all three cases and there is little or no difference between the Transverse and Longitudinal fracture case behaviour.

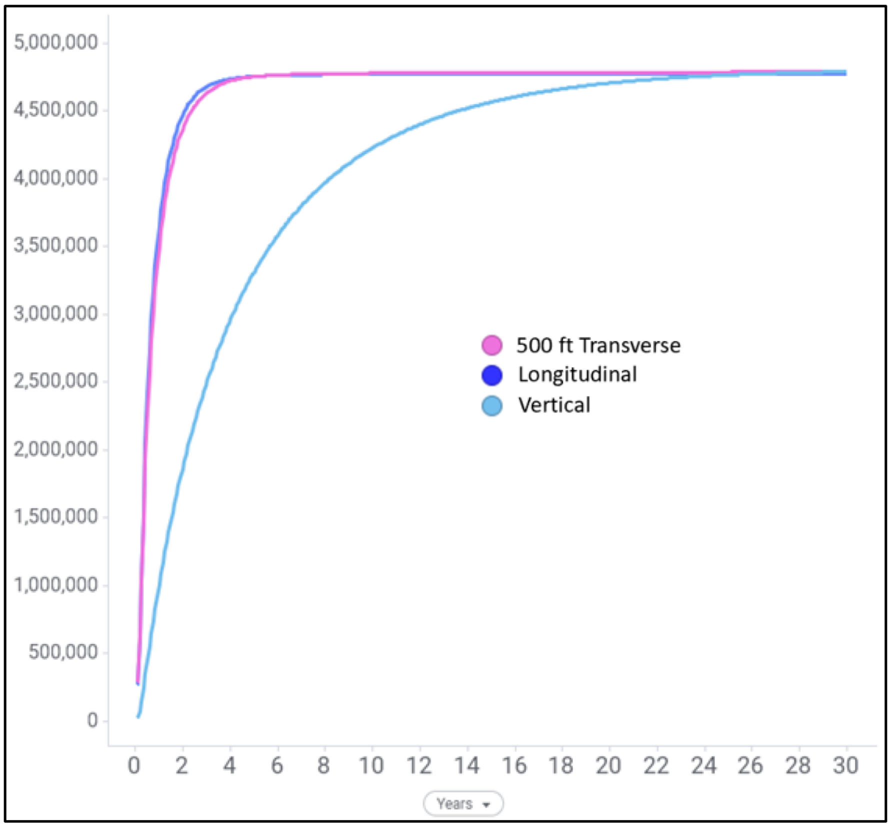

Figure 9—30-Year Cumulative Production for the 45 mD Case, with the Differing Completion Designs

Additionally, due to the relatively high reservoir permeability, the efficacy of all three well designs means that the reservoir block is fully drained and drawn down to the minimum bottomhole flowing pressure of ~1,000 psi after 30 years. Clearly there is a significant rate acceleration in both of the horizontal well designs, as a result of the greater initial/early-time reach from the wellbore from the ten fracture stages of the transverse well or the fully stimulated longitudinal design. However, cost considerations of the vertical vs. the horizontal well cases would also need to be considered.

For this moderate 45 mD permeability case, there is little benefit to increasing the frac count per stage as shown in Figure 10.

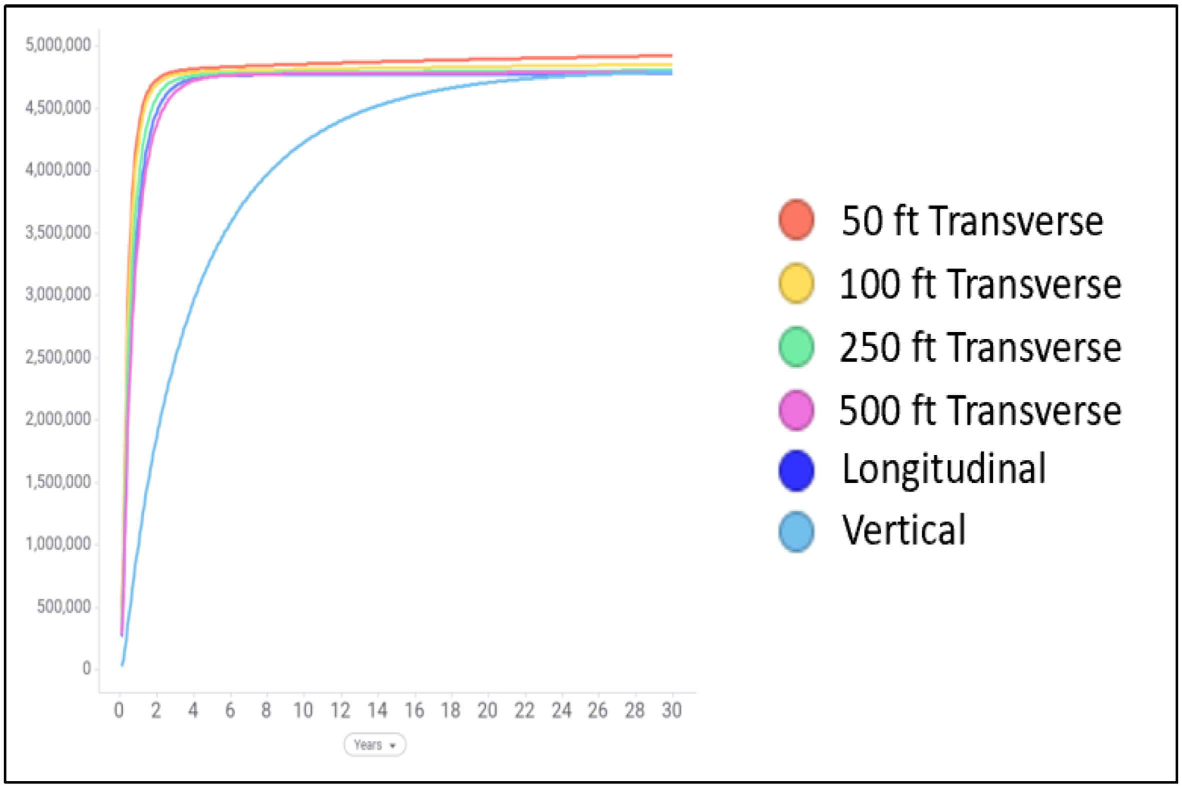

Figure 10—30-Year Cumulative Production for the 45 mD Case, With Multiple Transverse Fractures per Stage

It is somewhat intuitive, and indeed in agreement with broad industry experience, from the development of moderate permeability reservoirs, that an effective depletion can occur with the use of either multiple vertical wells with single stage fracture treatments, or horizontal wells with relatively large fracture spacing. As has been modelled in this example with 500 ft fracture spacing, though in reality the most effective fracture (and well) spacing, at this permeability, could be significantly larger, due to the ability to draw down the formation for >2000 ft over a 30-year time period as demonstrated in the vertical well design case. Such additional sensitivities should always be performed in order to optimize an individual case.

However, from the past twenty years of experience in developing unconventional formations in North America we know that there can be significant benefit to decreasing the fracture spacing in unconventional permeability reservoirs. Therefore, the fundamental question that we need to address is to what degree of reservoir permeability can we extend the moderate permeability, high fracture conductivity development approach to increasingly lower permeability formations while perhaps at the same time embracing the application of a tighter fracture spacing?

In an attempt to provide some broad guidelines in answering this fundamental development question, we re-ran the simulations for all three of the development cases, together with a variation in the fracture spacing for the Transverse case from 500 ft. to 50 ft. stage/frac spacing. The results of these simulations are provided in Figure 9 which shows the production profiles and drawdown pressures associated with the three well completion designs. These simulations were performed in reservoirs having permeabilities up to seven orders of magnitude lower – in reservoir permeabilities of 4.5 millidarcy; 450, 45 & 4.5 microdarcy; and 450 and 45 nanodarcy. (i.e. we ran the simulations across the full range of low permeability reservoirs down to the realm of single microdarcy and nanodarcy unconventional reservoirs).

Figure 12 provides a visualization of the 30-year reservoir pressure depletion as a function of permeability for the cases of the longitudinal fractured horizontal development and multiple transverse fractured horizontal well with 500 ft & 50 ft fracture spacing.

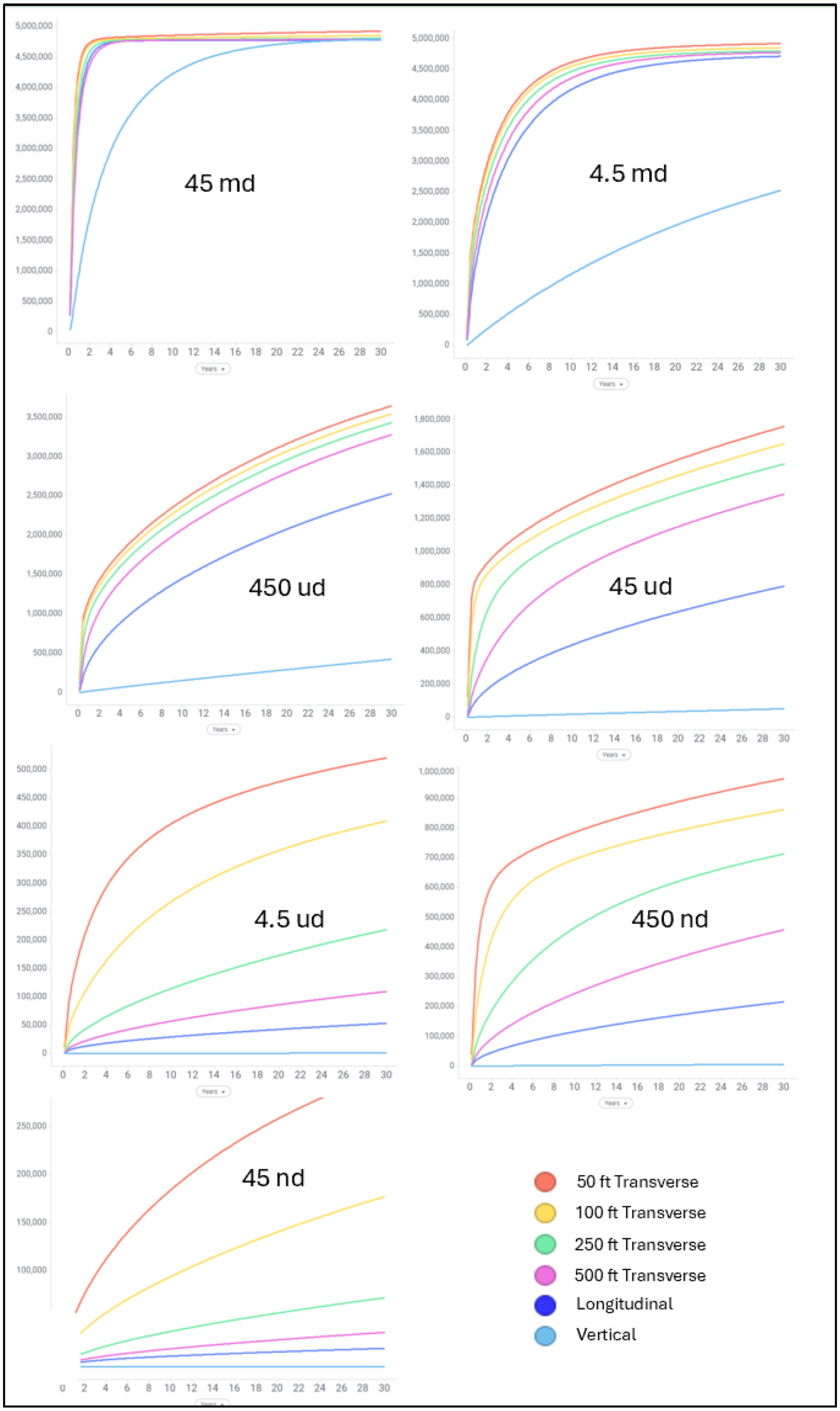

Figure 11—Production Performance of 3 Vertical & Horizontal Well Completion Designs in Low to Unconventional Permeability Formation.

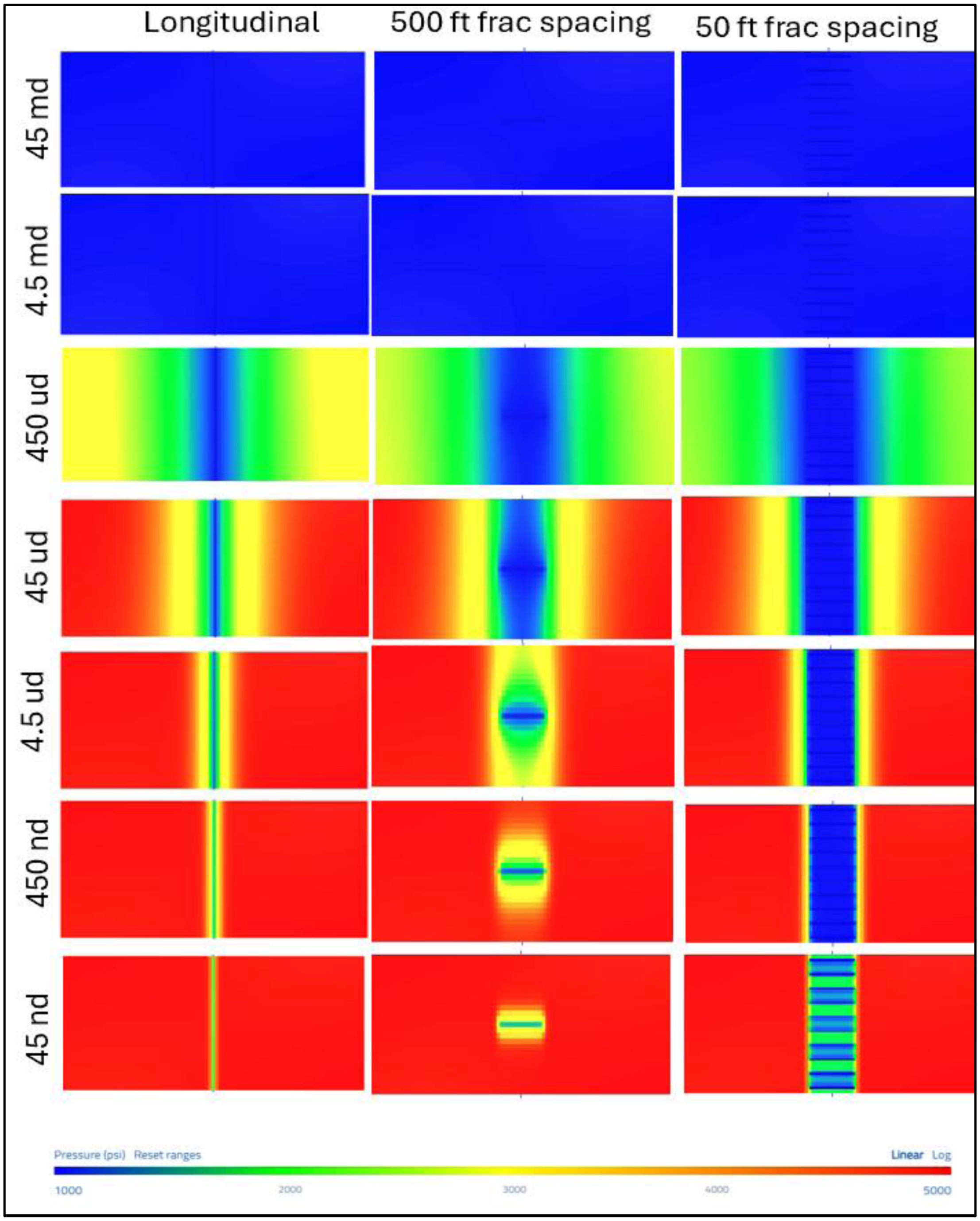

Figure 12—30-Year Reservoir Pressure Depletion Plots as a Function of Permeability for the Longitudinal and 500 ft & 50 ft Multiple Transverse Fracture Horizontal Well Designs.

As would be expected, as the permeability continues to decrease there is a benefit to using a Transverse fractured horizontal well approach – and the increased reservoir contact area from decreasing the fracture spacing. This can clearly be seen in Figure 13 (Left) and Figure 13 (Right), which show the production and recovery per well, for all permeability and Transverse fracture densities investigated.

In unconventional permeability reservoirs of 4.5 microdarcy and lower there is clearly a much greater productivity benefit to having the fractures spaced closer together. Conversely, at reservoir permeabilities of 450 microdarcy and above there is very little benefit to decreasing the fracture interval spacing.

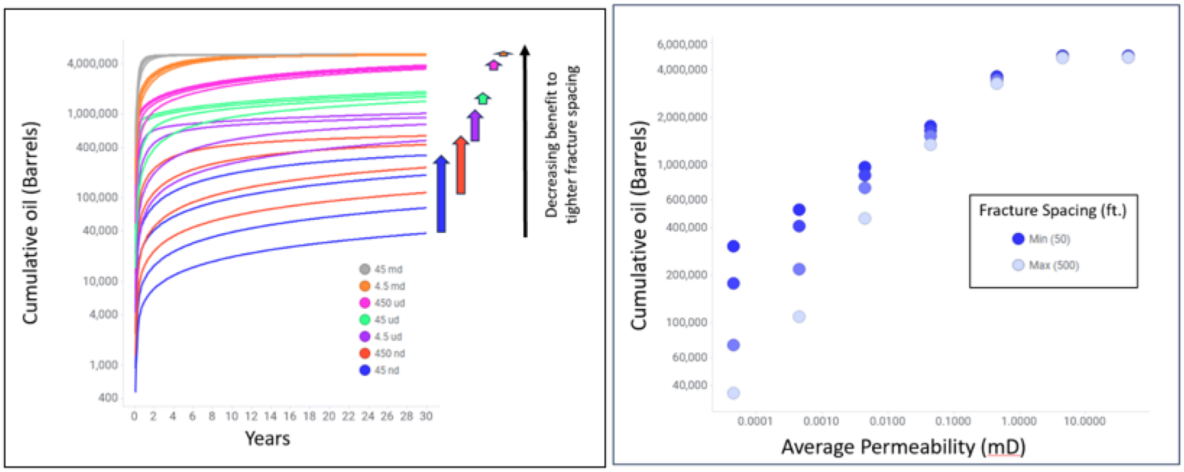

Figure 13—Production Performance as a Function of Fracture Spacing vs. time (Left) and vs. a Range of Permeabilities (Right).

FIELD DEVELOPMENT COST COMPARISON

Field development(s) can be an extremely capital-intensive exercise to perform. Most Operators will typically assign all of the associated development costs to an individual Well AFE (Authorization For Expenditure) assessment, which is then typically routed for both internal approval and external approval from Partners and Participants in the well / development.

In the Part II paper, the authors presented the total well costs of a typical unconventional 10,000 ft lateral Bakken well with separate Drilling, Completions, Downhole Production, and Surface Pad & Facility components. Such an AFE format is shown for a 2024 vintage, 2-mile lateral horizontal well drilled by Liberty Resources in the first two columns of Table 3. The total Unconventional Well AFE of 7.36 US $million may be divided into a Drilling component of 2.5 US$million, a Completions component of 2.84 US $million, a Downhole Production equipment costs of $ 160 K.USD, and finally a Surface Pad & Facilities of 1.86 US$million. The completion in this case was for a low proppant loading (1,000 lb/ft), slickwater fracture stimulation case using low-cost white sand. For this Part III paper as we are considering a 5,000 ft lateral well with a high-permeability ceramic proppant with a 2 lb/ft^2 design placed with a crosslinked gel fluid system. These cost changes are calculated below in Table 3 and Table 4 respectively to provide a variety of well costs from 7.57 US$million for the 500 ft transverse (or longitudinal) horizontal case, to 21.22 US$million if the stimulation is increased to 50 ft separation between the fractures.

Table 3—Unconventional Bakken Well Costs.

BAKKEN UNCONVENTIONAL WELL COSTS	10,000 ft Lateral Unconventional (Slickwater) Actual Well Costs (US$million)	5,000 ft Lateral Unconventional (Slickwater) Actual Well Costs (US$million)	Vertical Unconventional (Slickwater) Calc. Well Costs (US$million)
Drilling	2.500	2.500	1.900
Completion	2.840	1.846	0.500
Production (Downhole)	0.160	0.160	0.160
Pad & Facilities	1.860	1.860	0.990
TOTAL COST (US$million)	7.360	6.366	3.550

Table 4—Calculated Bakken Transverse 5,000 ft Lateral Well Cost as a Function of Fracture Spacing.

	BAKKEN UNCONVENTIONAL ACTUAL WELL COSTS	BAKKEN UNCONVENTIONAL EQUIVALENT WELL COSTS (for X-Link Gel /High Cond. Propped Stimulation) WITH VARIOUS FRAC SPACING			
COMPLETION COSTS	**Frac Spacing (ft)**	**50**	**100**	**250**	**500**
Pounds of Proppant / ft of Lateral	1,000	10,000	5,000	1,800	900
Incremental Proppant Costs (US$million))	0.00	14.85	7.35	2.55	1.20
Total 5,000 ft^2 Cost (US$million)	1.85	16.70	9.20	4.40	3.05
TOTAL 5,000 ft LATERAL COSTS					
Drilling	2.50	2.50	2.50	2.50	2.50
Completions	1.85	16.70	9.20	4.40	3.05
Production (downhole)	0.16	0.16	0.16	0.16	0.16
Pad & Facilities	1.86	1.86	1.86	1.86	1.86
TOTAL WELL COST (US$million)	6.37	21.22	13.72	8.92	7.57

Finally, Table 5 gives estimated well costs for Longitudinal or 500 ft Transverse fractured horizontal wells in differing geographical regions using the same cost multipliers from the Part II paper used to evaluate efficacy of unconventional horizontal well designs.

Table 5—Calculated Well Costs for Permian, Gulf of America (GOM), Rest of World (ROW) Onshore, and ROW Offshore.

	BAKKEN CONVENTIONAL (X-Link/LWC) CALCULATED WELL COSTS (US$million))	PERMIAN Multiplier	PERMIAN COSTS (US$million)	GOM Multiplier	GOM COSTS (MM.USD)	ROW ONSHORE Multiplier	ROW ONSHORE COSTS (US$million))	ROW OFFSHORE Multiplier	ROW OFFSHORE COSTS (US$million))
Horizontal (5,000 ft w/ 500ft Frac Stages)									
Drilling	2.500	1.15	2.875	1.50	4.312	1.50	4.312	1.50	6.468
Completion	3.050	1.15	3.503	2.00	7.006	2.00	6.532	1.50	10.509
Production (downhole)	0.160	1.15	0.184	1.50	0.276	1.50	0.276	1.50	0.414
Pad & Facilities	1.860	1.15	2.139	5.00	10.695	2.00	4.278	1.50	16.042
TOTAL (US$million)	7.570		8.70		22.29		15.40		33.43
Vertical Well									
Drilling	1.900	1.00	1.900	1.50	2.850	1.50	2.850	1.50	4.275
Completion (Single-Stage 450 Mlbs LWC)	0.700	1.00	0.700	2.00	1.400	3.00	1.500	1.50	2.100
Production (downhole)	0.160	1.00	0.160	1.50	0.240	1.50	0.240	1.50	0.360
Pad & Facilities (no high rate lift – 4 wells)	0.990	1.00	0.990	5.00	4.950	2.00	1.980	1.50	7.425
TOTAL (US$million)	3.750		3.750		9.440		6.570		14.160

The authors recognize that we have taken a broad estimate of costs by region – which may or may not reflect any individual location – but we believe the stated multipliers are a reasonable starting point. Using these data, and assuming a 35% Royalty burden, in Table 6 we compare Unit Development Cost (UDC) across regions and ranges of permeability for 'Conventional' style horizontal well fracturing designs. UDC is given by region for development of a 1-mile x 1-mile (640-acre) spacing unit using either a single vertical well or one horizontal well employing a 500 ft. fracture spacing completion design.

Table 6—Calculated Unit Development Costs for Conventional Horizontal Completions.

		BAKKEN		PERMIAN		GOM		ROW ONSHORE		ROW OFFSHORE	
CASE	TOTAL DSU OIL (BO)	DEVELOPMENT COST (US$million)	CALCULATED UDC ($/BO)	DEVELOPMENT COST (US$million)	CALCULATED UDC ($/BO)	DEVELOPMENT COST (US$million)	CALCULATED UDC ($/BO)	DEVELOPMENT COST (US$million)	CALCULATED UDC ($/BO))	DEVELOPMENT COST (US$million)	CALCULATED UDC ($/BO)
Horizontal – 45 mD	4,790,141	7.57	2.43	8.70	2.79	22.29	7.16	15.40	4.95	33.43	10.74
Horizontal – 4.5 mD	4,771,476	7.57	2.44	8.70	2.81	22.29	7.19	15.40	4.97	33.43	10.78
Horizontal – 450 μD	3,277,239	7.57	3.65	8.70	4.08	22.29	10.46	15.40	7.23	33.43	15.70
Horizontal – 45 μD	1,345,707	7.57	8.65	8.70	9.95	22.29	25.48	15.40	17.61	33.43	38.22
Horizontal – 4.5 μD	458,170	7.57	25.41	8.70	29.22	22.29	74.84	15.40	51.71	33.43	NA
Horizontal – 450 nD	109,150	7.57	NA	8.70	NA	22.29	NA	15.40	NA	33.43	NA
Horizontal – 45 nD	35,697	7.57	NA	8.70	NA	22.29	NA	15.40	NA	33.43	NA
Vertical – 45 mD	4,792,017	3.75	1.20	3.75	1.20	9.44	3.03	6.57	2.11	14.16	4.55
Vertical – 4.5 mD	2,524,904	3.75	2.28	3.75	2.28	9.44	5.75	6.57	4.00	14.16	8.63
Vertical – 450 μD	422,522	3.75	13.65	3.75	13.65	9.44	34.37	6.57	23.92	14.16	51.55
Vertical – 45 μD	51,361	3.75	NA	3.75	NA	9.44	NA	6.57	NA	14.16	NA
Vertical – 4.5 μD	6,037	3.75	NA	3.75	NA	9.44	NA	6.57	NA	14.16	NA
Vertical – 450 nD	663	3.75	NA	3.75	NA	9.44	NA	6.57	NA	14.16	NA
Vertical – 45 nD	71	3.75	NA	3.75	NA	9.44	NA	6.57	NA	14.16	NA

NA – Not Applicable – Greater than $ 100 per net BO UDC.

In the above Table 6 we have highlighted those cases where the Unit Development Costs are below $ 15 per BO. With current commodity pricing, this is a likely cut-off for so-called half-cycle development costs which only include well costs (so called DC&F, Drilling, Completion & Facility costs) without consideration of the additional costs of leasehold/property acquisition, exploration/appraisal, Gathering & Processing, or Overhead (General & Administrative) costs. The table above shows that a transverse conventional horizontal well completion has potential application in low permeability reservoirs. The UDC's are single digit to low teen values for the 450 microdarcy cases and higher permeabilities, in all but the very highest cost environment of the international offshore scenario.

Table 7 are the UDC costs as presented in the Part II paper for an "Unconventional" horizontal stimulation design. Comparison of the UDC data in Table 6 and Table 7 suggests that there is significant area of overlap in performance between a Conventional and an Unconventional completion approach to the development of lower permeability reservoirs, that falls somewhere within the range of ca. 0.5 to ~5 mD. Over this range, it is therefore much more likely that performance will consequently be more of a function of the well construction and service cost environment in which the Development takes place. Therefore, in a low cost, onshore North American location the cost efficacy of large unconventional stimulations provide lower UDC's – provided it is possible to actually place the slickwater completion as the permeability increases. At some level of inherent permeability – wherever it may exist onshore North America – it would be necessary to pivot to the pumping of more conventional fracture treatments where those characteristics are required. These factors might include the viscosity of say more impactful fluids such as cross-linked gels, in order to provide the necessary fluid leak-off control or indeed more permeable proppant in order to increase the FRR and reduce the required well-spacing based on the Local market and cost-conditions.

Table 7—Calculated Unit Development Costs for Unconventional Horizontal Well Completions (Pearson *et al*., 2025).

Case	TOTAL DSU OIL (BO)	BAKKEN	PERMIAN	GOM	ROW ONSHORE	ROW OFFSHORE
		CALCULATED UDC ($/BO)	CALCULATED UDC ($/BO)	CALCULATED UDC ($/BO)	CALCULATED UDC ($/BO)	CALCULATED UDC ($/BO)
Horizontal – 0.006 mD	3,397,991	13.33	15.33	39.51	27.89	59.26
Horizontal – 0.06 mD	6,885,936	6.58	7.56	19.50	13.76	29.24
Horizontal – 0.6 mD	12,452,539	3.64	4.18	10.78	7.61	16.17
Horizontal = 6.0 mD	25,146,525	1.80	2.07	5.34	3.77	8.01
Horizontal – 60.0 mD	30,515,061	1.48	1.71	4.40	3.11	6.60
Vertical 160 acre – 0.6 mD	2,320,803	18.83	18.83	47.94	34.84	71.91
Vertical 160 acre – 6.0 mD	8,211,561	5.32	5.32	13.55	9.85	20.32
Vertical 160 acre 60.0 mD	20,895,019	2.09	2.09	5.32	3.87	7.99

In both Offshore and International locations, the simulation results outlined above suggest that the economic development of low permeability oil reservoirs with either vertical or ERD wells is limited to formations with permeabilities of 0.5 mD range and above. For reservoir permeabilities in the range of 0.05 to 0.5 mD either a conventional or unconventional multi-fractured horizontal well approach can be investigated, while at those permeabilities below ~ 0.05 mD (50 microdarcy) then it is most likely that an Unconventional Completions approach would be preferred. Once again the consideration of the larger drainage area and well-spacing will play a role as these will almost always be higher cost environments than are currently encountered in onshore NA; where well construction costs may be significantly different.

In all circumstances, it is appreciated that these are transient factors and that what lies beyond the capability of development today and remains a stranded resource may not be the case tomorrow, as both technologies and economic factors change.

CONCLUSIONS AND RECOMMENDATIONS

The combination of the hydraulic fracturing technique with horizontal drilling has delivered a revolution to the development of oil and gas formations. Initial field developments took a "conventional" approach of using cross-linked fluid designs with high conductivity proppant placed in a relatively small number of fracture stages that were typically less than 10 fracs per well. The onshore North American Shale Revolution has taken these technologies and instead of relying on the hydrocarbon molecule moving hundreds or thousands of feet through the reservoir to the frac, as it would in permeable rock; it has "taken the frac to the molecule" through the placement of a large number (several hundred) of fracs per well using high injection rate fracturing where the fluid velocity is used to carry a relatively low concentration of low quality sand proppant into each frac. Both techniques have their place and in the three parts of this series of papers we have tried to take a holistic approach to the development optionality facing any practitioner in the industry. After analysing a number of well configurations, and fracture densities, and using a $ 15 per BO cut-off for development economics our analysis of the varying approaches can be summarized by the economic ranges displayed in Figure 14 below.

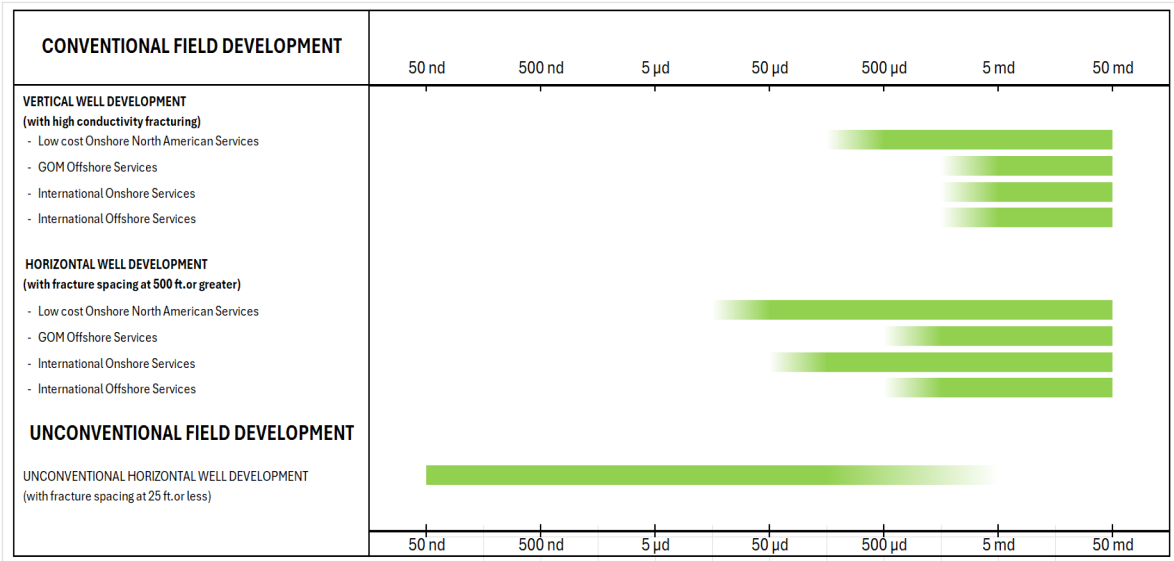

Figure 14—Permeability Ranges for Field Development Options Generating Unit Development Costs of < $15/BO.

For all regional service areas, we surmise that at those permeabilities below ~10 µD that the industry should always be using an Unconventional Horizontal Well approach to field development; and if well construction/service costs do not support that then there is very little that can be done (at this time). Development performance will be dominated and dictated by the well spacing, fracture density and the length of the horizontal lateral, and both the Conventional Vertical Well or Conventional Horizontal Well designs will not be economic under almost any circumstances. In the permeability range of ~10 µD to 500 µD, then it is likely that either a Conventional or an Unconventional completion approach to Horizontal Well development could be economic and should be worked up and considered. There may be specific drivers for one technique over another, that in differing applications may make one approach more appealing than another. At permeabilities between 0.5 mD (500 µD) and 5 mD all three approaches to development are valid options and the decision will likely be driven by regional capability and experience, the availability and cost of services (as well as at the upper end of this permeability range on the Unconventional fluid performance for fracture extension / proppant transport in the formation to be fractured etc..). Finally, for permeabilities above 5 mD, a Conventional approach to the development should always be pursued, although the choice of Vertical or Horizontal Well should be based on local considerations (e.g. in an offshore environment it is more common for horizontal developments due to the lack of extensive slot availability, and the same might be true in areas of environmental sensitivity or with other surface access difficulties).

It is envisaged, as more of these 'intermediate' permeability formations are developed, that the industry will garner experience, both good and bad, in determining the most applicable technique, for formation, contractual or environmental reasons. It is believed that the FRR, i.e. relative importance of the inter-frac drainage volume vs. the boundary drainage volume, will be one of the indicators that helps determine the most effective approach. As some of the unconventional basins begin to mature, surveillance programs and targeted pilot programs e.g. (McKimmy *et al.*, 2025) are demonstrating that effective drainage is a key measure to achieving success. In an environment where well cost(s) and industry limitations constrain well density/spacing, then high-intensity fracturing, is effectively over capitalising the wells. With fast intra-frac drainage, it is the boundary-drainage that is the long term deliverability system; and in these cases, the fracture effective-geometry and conductivity far outstrip the fracture density in terms of their relative importance. Fewer, higher quality fracs to transmit boundary flow to the wellbore is required. Conversely of course, where cost(s) allow high well-density, even at such low permeabilities, then intra-frac drainage is the dominant mechanism and we can afford to 'move the fractures to the molecule'.

The overlap permeability range from 10 μD to 500 μD is the most critical, and also the most uncertain region for tight-rock Development consideration. Over this sensitive range it is likely that its affordable well density that will dominate, and ultimately thereby the required balance of Compound-Drainage (intra-frac) and Boundary-Drainage.

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NOMENCLATURE

h	*Formation thickness (ft)*
k	*Formation permeability (mD)*
μ	*Fluid viscosity (cP)*
AFE	*Authorisation For Expenditure (mm.USD)*
ANS	*North Slope of Alaska*
BO	*Barrel of Oil (bbl)*
CT	*Coiled-Tubing*
ERD	*Extended Reach Drilling*
FCD	*Dimensionless Fracture Conductivity (dimensionless)*
FRR	*Fracture Recovery Ratio*
LWC	*Light Weight Ceramic*
NA	*North America*
NCS	*Norwegian Continental Shelf*
PPA	*Pounds Per gallon Added (lbs/gal)*
RCP	*Resin Coated Proppant*
SNS	*Southern North Sea*
UDC	*Unit Development Costs ($/BO)*

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